How Much Information Fits in a Vector?

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Abstract

Recent work in neural network interpretability has suggested that hidden activations of some deep models can be viewed as linear projections of much higher-dimensional vectors of sparse latent "features." In general, this kind of representation is known as a superposition code. This work presents an information-theoretic account of superposition codes in a setting applicable to interpretability. We show that, when the number k of active features is very small compared to the number N of total features, simple decoding methods currently used by sparse autoencoders are reliable if d is a constant factor greater than the Shannon limit. Specifically, when $\ln k / \ln N \leq \eta < 1$ and H is the entropy of the latent vector in bits, asymptotically it suffices that $d/H > C(\eta)$ for a certain increasing function $C(\eta)$. However, $C(\eta)$ varies significantly depending on the decoding method used. For example, when $\eta = 0.3$, we show that the popular method of top-k thresholding typically requires a factor of $C = 4$ dimensions per bit. On the other hand, we suggest a family of methods with comparable computational requirements (generalized matching pursuit) that only require $C < 2$ even as η approaches 0.4. Our results address a lack of practical references on superposition codes in a very sparse regime.

Introduction

If each neuron in given neural network coded for a "meaningful feature" of its input, we could hope to reverse-engineer the network's behavior on a neuron-by-neuron basis. However, many large models have been found to learn neurons that correlate simultaneously with apparently unrelated features. This phenomenon is known as polysemanticity. (See for example Nguyen et al. (2016), Zhang & Wang (2023) and Olah et al. (2020).)

The appearance of so-called "polysemantic neurons" is not surprising from a connectionist viewpoint. Since at least the 1980s, proponents of artificial neural networks have argued that these systems may naturally use **distributed representations**—coding schemes where individual features are represented by patterns spread over many units of computation, and conversely where each units carries information on many features. (This term was apparently coined in Rumelhart et al. (1986), Chapter 3.) In contrast, a *local*

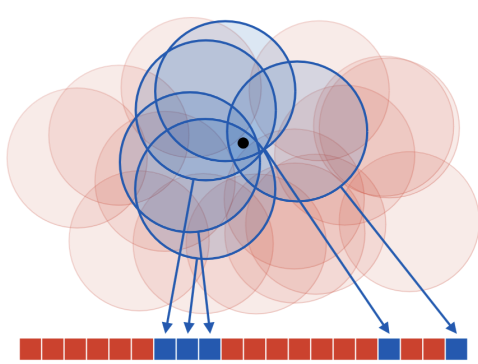


Figure 1: A coarse code representing a point on a plane. Each "neuron," drawn as a red or blue square, encodes whether the point belongs to an associated "receptive field." Although no neuron gives specific information on the position of the point, the overall code determines its position with reasonable accuracy.

representation would dedicate each unit to a single feature. See Thorpe (1989) for a general discussion of local and distributed codes. Figure 1 illustrates a classic example of a coarse code, one kind of distributed representation.

As of yet, relatively little is known about how large neural networks learn to represent information in their hidden layers or to what extent this information can be interpreted. However, should "interpretable features" exist, the connectivist viewpoint makes it natural that they would be stored with non-local codes. This is a common assumption in interpretability research today; for example, when Meng et al. (2022) intervened on an MLP layer of a language model to "edit" a factual association, both the "subject" and the "fact" were modeled as vectors of activations rather than individual channels.

To infer some kind of latent features from an activation vector y, one simple proposal is to model y as a linear projection $y = Fx$ of high-dimensional and *sparse* vector x. This idea is based on the remarkable fact that, for certain families of matrices F and for certain constraints on the sparsity of x, a d-dimensional linear image y codes uniquely for an N-dimensional sparse vector x when $N = \Omega(e^d)$.

We refer to the columns of F as codewords and the whole matrix F as a dictionary. Since y is a linear superposition of codewords, it is called a **superposition code** for x. The task of inferring x from a superposition code is known as sparse reconstruction, and the task of inferring the dictionary F from a distribution over the codes y is called dictionary learning. Both of these problems have been studied in the field of compressive sensing, although with different applications in mind. (See Elad (2010) for a review of classic work in the context of signal and image processing.)

Already in 2015, Faruqui et al. (2015) used a dictionary learning method to derive sparse representations for word embeddings and argued that these latents were more interpretable than the original embedding dimensions. More recently, a series of works beginning with Yun et al. (2021) have applied dictionary learning to the internal representations of transformer-based language models. Cunningham et al. (2023) suggested the use of **sparse autoencoders** (SAEs) and Templeton et al. (2024); Gao et al. (2024) scaled sparse autoencoders to production-size large language models.

Templeton et al. (2024) showed that latent features learned by SAEs are often highly intepretable, and that intervention on these features allows "steering" language models in predictable ways. However, as reported in Gao et al. (2024), even SAEs with extremely large numbers of latents suffer from an apparently irreducible reconstruction error. According to Sharkey et al. (2025), understanding the limitations of SAEs—and dictionary learning in general—is an important open question in the research program of mechanistic interpretability.

Contributions

To infer a latent representation $x \in \mathbb{R}^N$ from an activation vector $y \in \mathbb{R}^d$, sparse autoencoders use a simple estimate of the form $\hat{x}(y) = \sigma(Gx)$ where $G\colon \mathbb{R}^{N\times d}$ is a learnable matrix and σ is some kind of thresholding operation. Throughout this paper, we refer to any estimate of this form as a "one-step estimate" for x, since it requires only one step of computation. Gao et al. (2024), a work representative of research on large-scale sparse autoencoders, considered latent vectors with dimension N on the order of one million the number of non-zero coefficients k in each latent vector $\hat{x}$ was around one hundred.

Research on sparse autoencoders is mired by many kinds of scientific uncertainty. Of course, it is not known a priori how well activation vectors can be effectively modeled as sparse superposition codes. (It is not even necessarily clear what is meant by an "interpretable feature," as per (cite something)). Furthermore, even if some activity of a model can be modeled as a superposition code, it's not obvious whether the latent features being encoded can be recovered by a one-step estimate. Indeed, the compressive sensing literature provides many iterative methods for sparse recovery that require more computation but succeed more reliably. To what extent are the limitations of SAEs explained by the limitations of our sparse recovery methods?

One way to frame this question is to ask *how much information* a superposition code may hold. Classically, this is addressed by the tools of information and coding theory. Given a "channel" with certain characteristics—for example, a band-limited telephone connection or a binary storage device with a cer-

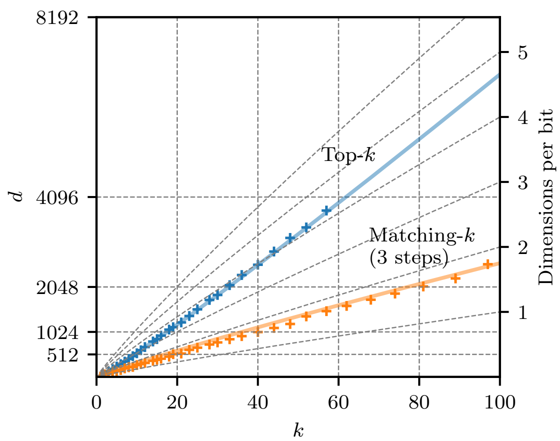


Figure 2: Crosses give embedding dimensions d required for two different methods to completely decode a superposition code for a k-sparse subset of $\{1, \ldots, 2^{20}\}$ with empirical probabiliy 0.95. and colored lines give rules of thumb derived from our theoretical discussion. The inverse "bitrate" $d/\tilde{H}_2$, where $\tilde{H}_2 = k \log_2(eN/k) \approx \log_2 \binom{N}{k}$, is indicated on the right axis. Top-k is a method currently used by sparse autoencoders, and three-step matching-k is a new method introduced in this work.

tain failure rate—the amount of information we can encode is asymptotically characterized by a channel "capacity" measured in bits per unit of channel usage.

The goal of this paper is to perform a similar analysis for sparse superposition codes in a setting that is meaningful to sparse autoencoders. More specifically, we consider a situation where the latent vector x is extremely sparse and must be recoverable from the code y by a relatively efficient method. Our analysis focuses on a specific toy example described in Section 1. Our main contributions are the following.

1. Asymptotically in a regime of sublinear sparsity, we prove that one-step estimates used by sparse autoencoders can reliably decode superposition codes $y \in \mathbb{R}^d$ so long as d is a constant factor greater than the Shannon limit (Proposition 4). One expression of our asymptotic result leads to an accurate rule of thumb for the empirical performance of the popular top-k method.
2. We investigate a simple extension of top-k which we call **generalized matching pursuit** (GMP). Numerical experiments show that, for the same values of (N, d), GMP can decode superposition codes carrying more than two times as much information while requiring only a small constant factor more computation than top-k.

A concrete result of our findings is illustrated in Figure 2. For $k \in \{1, \ldots, 100\}$ we show the number of dimensions d required for top-k and GMP to accurately read a k-sparse subset of $\{1, ..., 2^{20}\}$ that has been transmitted as a d-dimensional superposition. (The dictionary F is randomly generated; see Section 1 for more details.) As k increases, the ratio d/H of dimensions per bit of entropy required for top-k to perform reliably increases. For moderate values of k, top-k requires more than 4 dimensions per bit. In contrast, three-step matching-k requires only two dimensions per bit, while requiring only around 3 times more FLOPs for the same values of (N, d).

Although our analysis is limited to a simple example, our results suggest that

Related Work

Here: some discussion of the relation of our work with compressive sensing.

- Bajwa et al. (2010)
- Takeuchi (2024)
- Donoho et al. (2009)
- Donoho & Tanner (2009)

Structure of This Work

The following sections are structured as follows.

- Section 1 introduces the toy superposition code to be studied and the decoding methods we will consider. These are MAP threshold decoding, top-k decoding, and GMP.
- Section 2 briefly realls the idea of a Shannon limit and explains two different heuristic predictions for the number of dimensions that may be required for a superposition code to be decodable.
- Section 3 gives our main theoretical results on the performance of threshold and top-k decoding and our numerical experiments.
- Finally, Section 4 discusses our choice to use random dictionaries.

1 Superposition Codes and Sparse Recovery

We begin by describing the toy scenario to be studied. Given a large number N, consider a map F that represents each subset $x \subseteq [N] = \{1, \ldots, N\}$ by a linear combination

$$y(x) = \sum_{i \in x} f_i \in \mathbb{R}^d,$$

where the vectors $\{f_i \in \mathbb{R}^d : i \in [N]\}$ are chosen in advance and where the dimension d of the encoding is expected to be much smaller than N. The vectors f_i are called codewords for the elements of $[N]$, the collection $\{f_1, \ldots, f_N\}$ is called a dictionary, and the image Fx is called a superposition code. It is useful to view x as a vector in $\{0,1\}^N$ with coefficients

$$x_i = \begin{cases} 1 : i \in x \\ 0 : \text{otherwise} \end{cases}$$

and view F the matrix of column vectors $[f_1 \ \ldots \ f_N]$, so that $y = Fx$ is a linear equation. In this work, we'll model our subset as a random variable X uniformly distributed over all subsets of some fixed size $k \ll N$. Keeping with the orders of magnitude discussed in Gao et al. (2024), we consider $N \approx 1{,}000{,}000$ and $k \approx 100$.

The problem of recovering a sparse signal x from a linear projection Fx is known as sparse recovery. This problem is considered in the subject of compressive sensing. In this work, we will use language from coding theory and think of x as some data to be encoded and the projection $y = Fx$ as a code. Thus, sparse recovery becomes the task of "decoding" x. Note that this is opposite to the convention used in sparse autoencoders, where the map estimating x from y is called the "encoder." (In classical applications of autoencoders, the learned representation typically is understood as an efficient code for the data distribution, and typically has smaller dimension than the variable being modeled.)

Due to a connection with sparse linear regression, the subproblem of identifying the support of x from the projection Fx is often called model selection. When the coordinates of x take values in $\{0,1\}$, sparse recovery reduces to a problem of model selection. In general, note that model selection presents the main difficulty of sparse recovery; once the support of x identified, its non-zero coordinates can be estimated easily e.g. by solving a least squares problem.

In practice, a dictionary F employed by a neural network will likely have special properties related to the nature of the data being encoded and the operations that need to be performed. However, for the purposes of encoding a uniformly random subset of $[N]$, one natural way to choose a dictionary is to make each codeword an independent random vector. The simplest motivation for this strategy is the fact that, so long as $d = \Omega(\ln N)$, a dictionary of N randomly chosen vectors is highly "incoherent." For example, we can prove the following for dictionaries of Rademacher vectors.

Proposition 1. *For $\mu > 0$ and $p > 0$, let $d > 2\mu^{-2}(2\ln N + \ln p^{-1})$, and let $\{F_1, \ldots, F_N\}$ be independent draws from $\{-1/\sqrt{d}, 1/\sqrt{d}\}^d$. Then*

$$\forall i \neq j,\ |\langle F_i, F_j\rangle| < \mu \tag{1}$$

with probability at least $(1-p)$.

For a standard proof, see Section C.

The infimum of all constants μ satisfying the condition (1) is called the mutual coherence of a dictionary. Bounds on mutual coherence give a relatively simple way to guarantee the success of various sparse recovery algorithms. (For example, see Chapter 4 of Elad (2010).) In our scenario, we can say the following.

Remark 1. *Let $x \subseteq [N]$ be a k-sparse subset. Given a dictionary F with mutual coherence smaller than $1/2k$, for any k-sparse vector $x \in \mathbb{R}^N$ and for all $i = 1, \ldots, N$ we have*

$$\forall i, x_i = \begin{cases} 1 : \langle f_i, Fx\rangle \geq 1/2 \\ 0 : \textit{otherwise.} \end{cases}$$

Altogether we find that, for a fixed k, it suffices to take $d = \Omega(\ln N)$ in order for each coefficient x_i to be retrievable in a very simple way. In the remainder of this work, we are interested in refining this basic estimate.

We consider two natural random dictionary constructions. A **Rademacher dictionary** is a random dictionary with codewords drawn as in Proposition 1, i.e. independent Rademacher vectors normalized to have unit norm. A **spherical dictionary** is a random dictionary with codewords drawn uniformly and independently from the unit sphere. These dictionaries have similar incoherence properties in our regime of interest; a result we prove for one dictionary (such as Proposition 1) can always be proven in an equivalent form for the other.

Finally, it remains to define the decoding methods we will study. We informally refer to decoding methods that require only a single "step" of estimation and thresholding—like the estimate in Remark 1—as **one-step decoders**. In contrast, the field of compressive sensing provides a number of more iterative methods for sparse recovery, which are known to perform more reliably in general. However, many of these methods are. In this work, we consider a family of relatively cheap iterative algorithms which we call **generalized matching pursuit (GMP) decoders** .

1.1 Threshold Decoding and Top-k Decoding

To motivate one-step decoders, consider the problem of estimating single a coefficient X_i from the model

$$Y = X_i f_i + \overbrace{\sum_{j \neq i} X_j f_j}^{Z_i}. \tag{2}$$

(Here we write X and Y with capital letters to emphasize they are random variables.) If we approximate the sum Z_i as centered Gaussian noise with covariance Σ, the inference problem for X_i becomes tractable; specifically, if we define an inner product by $\langle v, w\rangle_\Sigma = \frac{1}{2} v^T \Sigma^{-1} w$, then the posterior on X_i can be expressed in terms of the linear function

$$\lambda_i(Y) = \frac{\langle f_i, Y\rangle_\Sigma}{\|f\|_\Sigma^2}.$$

In signal processing, this kind of optimal linear measurement is called a matched filter. In terms of the matched filter,

$$\ln \mathrm{P}(X_i = x | Y = y) = -\frac{\rho_i}{2}\left(x - \lambda_i(y)\right)^2 + \ln \mathrm{P}(X_i = x) + C \tag{3}$$

where $\rho_i = (\operatorname{Var} \lambda_i(Z_i))^{-1}$ is called the "signal-to-noise ratio." (See Section A for a review.)

When F is a random dictionary of unit-norm codewords, it's natural to approximate the covariance Σ by a factor of the identity, in which this case we have $\lambda_i(Y) = \langle f_i, Y\rangle / \|f_i\| = \langle f_i, Y\rangle$.

One natural simple decoding strategy is to base our estimate $\hat{X}_i$ directly on the matched filter $\langle f_i, Y\rangle$. Specifically, we can estimate $\hat{X}_i = 1$ if $\langle f_i, y\rangle$ is larger than some threshold τ and $\hat{X}_i = 0$ otherwise. We refer to this as **threshold decoding** at level τ (Algorithm 1). Substituting the prior

$$\mathrm{P}(X_i = 0) = \frac{N-k}{N}, \quad \mathrm{P}(X_i = 1) = \frac{k}{N}$$

into the approximate posterior (3) shows that the maximum a posteriori estimate for X_i corresponds to thresholding at level

$$\tau = \frac{1}{2} - \rho_i^{-1} \ln \frac{\mathrm{P}(X_i = 1)}{\mathrm{P}(X_i = 0)} = \frac{1}{2} - \rho_i^{-1} \ln \frac{k}{N-k}. \tag{4}$$

The inverse signal-to-noise ratios ρ_i^{-1} can be estimated by computing the variance under a random dictionary F. For both Rademacher and spherical dictionaries we have $\mathrm{Var}\,\langle F_i, F_j\rangle = 1/d$ for a pair of independent codewords, so

$$\rho_i^{-1} = \mathrm{Var}\langle F_i, Y\rangle = \mathrm{Var}\langle F_i, F_1 + \cdots + F_k\rangle = \begin{cases} (k-1)/d & : i \in X \\ k/d & : \text{otherwise.} \end{cases}$$

Since we do not know in advance whether $i \in X$, we take ρ to be the expected value

$$\begin{aligned} \rho^{-1} &= \mathrm{P}(X_i = 1)\,\mathrm{Var}(\langle F_i, Z_i\rangle | X_i = 1) + \mathrm{P}(X_i = 0)\,\mathrm{Var}(\langle F_i, Z_i\rangle | X_i = 1) \\ &= \frac{k(k-1) + (N-k)k}{Nd} = \frac{k - k/N}{d}. \end{aligned}$$

Putting this estimate for ρ_i^{-1} into (4) gives

$$\tau_{\mathrm{MAP}} = \frac{1}{2} - \frac{k - k/N}{d} \ln \frac{k}{N-k}.$$

We refer to threshold decoding using this value of τ as MAP threshold decoding. We can simplify this expression in the regime $k/N = o(1)$ by noting that

$$\tau_{\mathrm{MAP}} = \frac{1}{2} - \frac{k}{d} \ln \frac{k}{N} + O\left(\frac{k}{N}\right). \tag{5}$$

Another very simple strategy is to first compute all the matched filters $F^T Y$ and find the largest k coordinates. We will call this top-k decoding (Algorithm 2). Top-k has the disadvantage that k must be known exactly in advance, whereas in threshold decoding only requires an estimate of k to adjust the threshold τ. However, should k be known, it is clear that top-k is strictly more reliable than threshold decoding for any τ.

Input: $y \in \mathbb{R}^d$, $F \in \mathbb{R}^{d\times N}$, $\tau \in \mathbb{R}$
Output: $\hat{x} \in \{0,1\}^N$
$\hat{x} \leftarrow 0$;
$\hat{x} \leftarrow 1$ where $F^T y \geq \tau$;
return $\hat{x}$

Algorithm 1: Threshold decoding

Input: $y \in \mathbb{R}^d$, $F \in \mathbb{R}^{d\times N}$, $k \in \mathbb{N}$
Output: $\hat{x} \in \{0,1\}^N$
$\hat{x} \leftarrow 0$;
$\hat{x} \leftarrow 1$ at k largest values of $F^T y$;
return $\hat{x}$

Algorithm 2: Top-k decoding

Note that in some works (Elad (2010)) top-k is known as the thresholding algorithm. In sparse autoencoders, top-k is

1.2 GMP Decoding

Here: briefly introduce/motivate GMP decoding, compare to GOMP Wang et al. (2012).

Input: Code $y \in \mathbb{R}^d$, dictionary $F \in \mathbb{R}^{d \times N}$, sparsity k, steps T
Output: Latent $\hat{x} \in \{0,1\}^N$
Choose any step sizes $k_1, \dots, k_T$ so that $\sum_t k_t = k$ and $\max_t k_t$ is minimized ;
$\hat{x} \leftarrow \mathbf{0}$;
for $t = 1, \dots, T$ **do**
  $\hat{x} \leftarrow 1$ at k_t largest values of $F^T(y - F\hat{x})$;
end
return $\hat{x}$

Algorithm 3: Generalized matching pursuit (GMP) with T steps

2 Information Theory Bounds

To understand the minimum dimension d required for a superposition code with parameters (N, k) to be decodable by some method, it will be helpful to use the point of view of information theory.

Let us first consider the case $k = 1$, meaning that the latent variable X to be encoded is a just symbol drawn from an alphabet of size N. What is the minimum dimension d such that each value of X can be represented by a vector $Y \in \mathbb{R}^d$?

When the vector Y is be stored exactly, the answer to our question simply depends on the number of values it can take. If the coefficients of Y are 16 bit floating point numbers, then each coefficient can store (nearly) 2^{16} distinct numbers, and overall we need only about $d_{\min} \approx \frac{1}{16} \log_2 N$ dimensions. However, suppose that the coordinates of X are not stored exactly due to the presence of random interference. Characterizing the "capacity" of a vector in the presence of interference is a basic problem in coding theory.

One common model is to consider that X is subject to additive white Gaussian noise. More specifically, we let $Z \in \mathbb{R}^d$ be a vector of independent Gaussians each of variance P and consider the problem of recovering Y from $X + Z$. The following is then a typical (but remarkable) result of information theory.

Proposition 2 (A Shannon limit)**.** *Let the random variable $X \in [N]$ be uniformly distributed and let $Z \in \mathbb{R}^d$ be independent Gaussian noise of variance P. Suppose there exists a pair of maps $F\colon [N] \to \mathbb{R}^d$ and $G\colon \mathbb{R}^d \to [N]$ so that*

$$G(F(X) + Z) = X$$

with probability at least $(1-p)$, and suppose the variance of each coordinate of $F(X)$ is bounded by 1. Then

$$d \geq 2 \frac{(1-p)\ln N - \ln 2}{\ln\bigl(1 + P^{-1}\bigr)}.$$

Asymptotically for large N, this means that

$$\frac{d}{H} \geq (1 + o(1)) \frac{2}{\ln(1 + P^{-1})}$$

if there is any pair (F, G) that transmits X with probability greater than 0, where $H = \ln N$ is the entropy of X. In other words, transmitting at least $2/\ln(1 + P^{-1})$ dimensions per nat.

Conversely, it can be shown that our bound on the ratio d/H is asymptotically tight. More specifically, for any $\epsilon > 0$ and for sufficiently large N, it is enough to take

$$\frac{d}{H} \geq (1 + \epsilon) \frac{2}{\ln(1 + P^{-1})}$$

for there to exist some pair (F, G) satisfying the conditions of Proposition 2 with p arbitrary close to 0. For a reference on this fact, see Chapter 9 of Cover & Thomas (2006).

In the remainder of this work, we will apply a similar point of view to our toy superposition coding problem. Specifically, we will understand the performance of a given decoding method in terms of the attainable inverse "bitrate" d/H, where d is the dimension of our superposition code and $H = \ln \binom{N}{k}$ is the entropy of the k-element subset to be transmitted.

Ideally, it would be possible to decode a superposition code so long as d/H is larger than some fixed lower bound. Informally, this means that the decoder is able to read a non-vanishing amount of information from each dimension of the code. Since $\ln \binom{N}{k} = k \ln(N/k) + O(k) \leq k \ln(eN/k)$ (see Section B) in general this would mean having a sufficient condition of the form

$$d \geq Ck \ln(eN/k)$$

for some constant C. In fact, this kind of condition is proven in compressive sensing for other sparse reconstruction methods (see Inequality (1.2) in Donoho & Tanner (2009).)

How large do we expect the constant C needs to be? (In other words, how much information can we expect to read from each coordinate of a superposition code?) In the case $k = 1$, the top-k decoder succeeds so long as the inner product between each pair of codewords is numerically distinguishable from 1. It would be enough to simply enumerate (CONTINUE HERE)

> Here: remark on why the constant C is likely not too small and that we should really have
>
> $$d \geq k \ln N.$$

3 Main Results

For a given family of dictionaries $F_{N,d} \in \mathbb{R}^{N \times d}$, and consider the problem of recovering a random subset k-element subset X from a d-dimensional superposition code. Let

$$b(d, k, N; \tau), \quad b(d, k, N; \text{top-}k)$$

denote the respective failure probabilities for threshold decoding at level τ and for top-k decoding.[1] Naturally, we have

$$b(d, k, N; \text{top-}k) \leq b(d, k, N; \tau),$$

since if Y is decoded by threshold decoding at any level, it is also decodable by top-k.

Our first result provides an asymptotic constraint on the dimension d required for either of these values to be converge to 0 for large N.

Proposition 3. *Let $\epsilon > 0$ be arbitrary. Over the regime*

$$d \leq (1 - \epsilon) 2k \ln N, \quad 2 \leq k < N/2,$$

for random spherical dictionaries it holds that

$$\lim_{N \to \infty} b(d, k, N; \textit{top-}k) = 1.$$

Now, consider this result from an information-theoretic view. In a regime when $k \geq \epsilon N$ for some $\epsilon > 0$, called a regime of linear sparsity, the entropy of X is

$$H = \ln \binom{N}{k} \geq k \ln \left(\frac{N}{k} \right) = O(N).$$

However, Proposition 3 shows that we need at least $d = \Omega(k \ln N) = \Omega(N \ln N)$ dimensions for top-k to read a superposition code, which is asymptotically larger. Informally, the amount of information that top-k can read from each dimension of the code converges to 0.

[1] The letter b is a mnemonic for the "bad event."

Is it natural to ask whether, in some asymptotic regime, a one-step decoder can read a non-vanishing amount of information from each dimension of a superposition code. In this situation, we will say the decoder is "information-efficient." By Proposition 3, we would need $H/k \ln N$ to be bounded below. Since $\ln \binom{N}{k} = k \ln(N/k) + O(k)$, as $N \to \infty$ we have

$$\frac{H}{k \ln N} = \frac{k \ln N - k \ln k + O(k)}{k \ln N} = 1 - \frac{\ln N}{\ln k} + O\left(\frac{1}{\ln N}\right).$$

So, one-step decoders can only be information-efficient in the regime $k \leq N^\eta$ for $\eta < 1$. In compressive sensing, this is known as the sublinear sparsity regime. The following

Proposition 4. *For $\eta \in (0, 1)$, let $k \leq N^\eta$. Then for any $\epsilon > 0$, over the regime*

$$d \geq (1 + \epsilon) 2 (1 + \sqrt{\eta})^2 k \ln N,$$

for Rademacher dictionaries it holds that

$$\lim_{N \to \infty} b(d, k, N; \tau(\eta)) = 0$$

where $\tau(\eta) = (1 + \sqrt{\eta})^{-1}$.

For $\eta < 2$, it can be shown that

$$\binom{N}{k} =$$

Since $2 + 4\sqrt{\eta} + 2\eta \leq 4 + 4\eta$, when $\eta = \ln k / \ln N$ we have

$$(2 + 4\sqrt{\eta} + 2\eta) k \ln N \leq (4 + 4\eta) k \ln N = 4k \ln(Nk)$$

which is a simple expression for the prediction of Proposition 4.

This result has a remarkable interpretation in terms of effective "channel capacity."

4 Are Random Dictionaries Optimal?

In the last section, we considered the performance of different decoding methods in a scenario where the dictionary F was randomly generated. Of course, the dictionaries employed by neural networks may have special structure. It is natural to ask whether our findings would be significantly different if the dictionary were specially designed in some way.

One natural measurement for the coherence of a dictionary is the mean squared interference of codewords

$$\xi(F) = \binom{N}{2}^{-1} \sum_{i<j} \langle f_i, f_j \rangle^2.$$

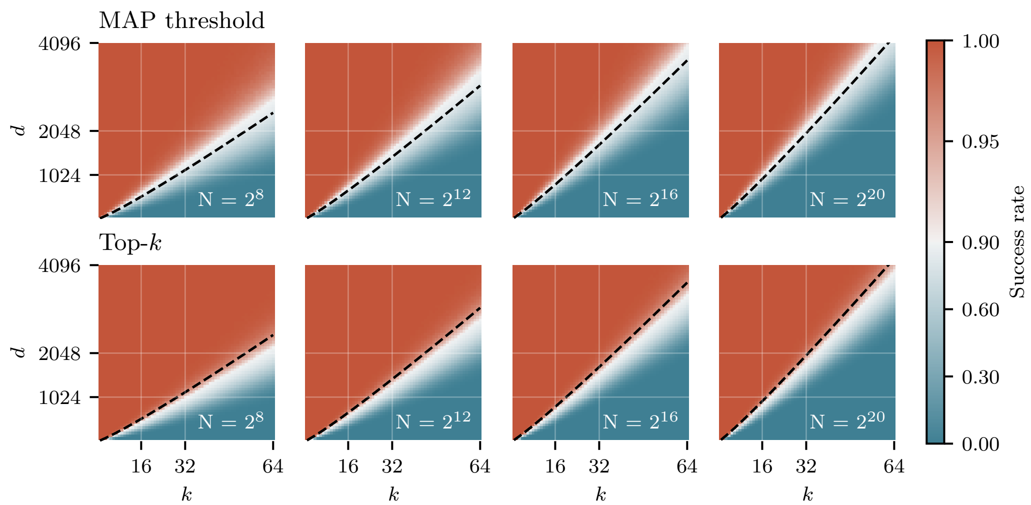
An important result in coding theory known as the Welch bound (Welch (1974)) states that, for any dictionary $F \in \mathbb{R}^{d \times N}$ of unit norm codewords, we have

$$\sum_{i,j} \langle f_i, f_j \rangle^2 \geq \frac{N^2}{d}.$$

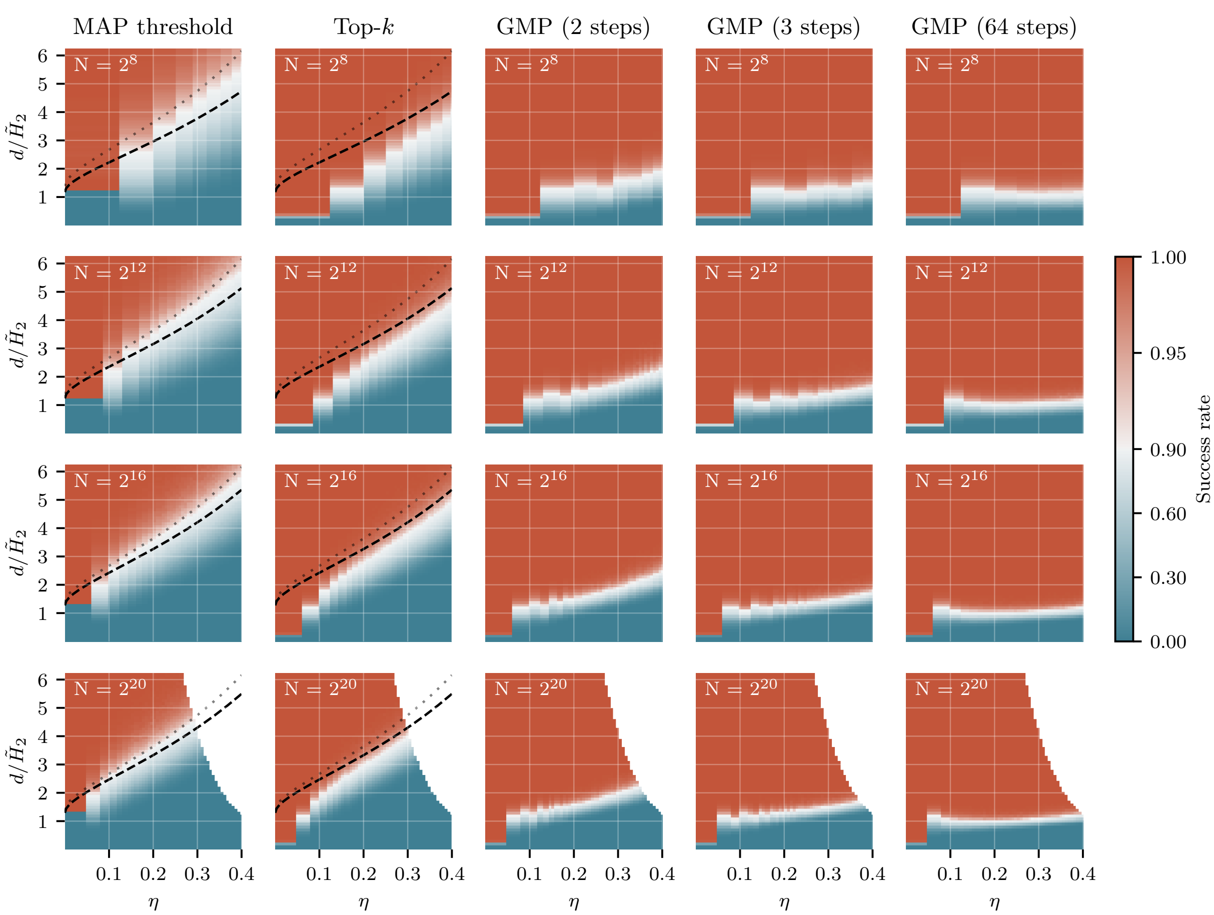
This can be derived by viewing the sum of squared inner products above as the sum of squared eigenvalues λ_i^2 of the Gram matrix $F^T F$ and applying the inequality $\sum_i \lambda_i^2 \geq N \sum_i \lambda_i$.

In terms of mean squared interference $\xi(F)$, the Welch bound means that

$$\xi(F) \geq \frac{1}{N^2 - N}\left(\frac{N^2}{d} - N\right) = \frac{1}{d}\left(1 - \frac{d}{N}\right)\left(1 + \frac{1}{N-1}\right).$$



(a) Empirical performance of MAP thresholding and top-k thresholding at decoding a k-element subset of $[N]$ from a d-dimensional superposition code using a random spherical dictionary. The bold line shows the critical value $d = 2(1+\sqrt{\eta})^2 k \ln N$ predicted for threshold decoding by Proposition 4.



(b) Empirical preformance of MAP thresholding, top-k, and GMP at recovering a superposition code using a random spherical dictionary, graphed as a function of inverse bitrate $d/\tilde{H}_2 = d/(k \log_2(eN/k))$ and the sparsity exponent $\eta = \ln k / \ln N$. The bold line shows the critical value $d = 2(1+\sqrt{\eta})^2 k \ln N$ predicted by Proposition 4 and the shows its limit as $N \to \infty$.

Therefore, in a limit where $N \to \infty$ and $d/N \to 0$, we have $\xi(F) \geq 1/d + o(1)$. However, recall that for a pair (F_i, F_j) of independent random unit-norm codewords we have

$$\mathrm{E}\langle F_i, F_j \rangle^2 = \frac{1}{d}.$$

(This is true for draws from the unit sphere, for normalized Rademacher vectors, etc..) Therefore, $\mathrm{E}\,\xi(F) = 1/d$ for a dictionary of random codewords. In the sublinear regime $k \leq N^\eta$ for $\eta < 1$, the previous section showed that one-step decoders are viable in regimes where $d/N \to 0$, so in this situation we conclude that mean squared inference $\xi(F)$ cannot be significantly improved.

Another natural measurement of coherence is the *maximum* absolute value of any inner product.

Here:

To validate our

5 Conclusions and Future Work

Over the course of computation, it is common for programs to use more memory than was required to store their input data. Similarly, it is natural to expect that the entropy of a "simple" (or "interpretable") description for the residual vectors within a language model is not bounded by the entropy of language. Methods like sparse autoencoders aim to provide descriptions of these hidden activations, but relatively little is known about the nature of the information is being carried. One basic question is *how much* information can hypothetically be stored in an activation vector using a superposition code.

Previous work showed that sparse autoencoders can help learn interpretable representations of the activity inside a neural network. However, the success of these methods is limited for reasons that are not yet well understood.

In this work, we have identified one point of view that might explain their limited success. In a toy scenario, we showed that the simple estimates these models use to infer sparse representations are much less "efficient," in an information-theoretic sense, than a simple iterative method. This is true even when the signal to be inferred is extremely sparse. To our knowledge, this kind of explicit, non-asymptotic comparison was not previously available in the literature.

Of course, we do not suggest that the latent signal stored by a typical neural representation is well-modeled as a uniformly random k-sparse subset. However, the "bitrate gap" between one-step estimates and matching pursuit opens a natural question: how much information can neural networks typically encode in their internal activity? Can they, like matching pursuit, read around one bit of mutual information from each neuron? If they can, our findings suggest that sparse autoencoders may be fundamentally unable to decode their representations. Overall, we hope the point of view of coding efficiency helps guide the interpretation of neural representations in future work.

References

Waheed U. Bajwa, Robert Calderbank, and Sina Jafarpour. Why Gabor frames? Two fundamental measures of coherence and their role in model selection. *Journal of Communications and Networks*, 12(4):289–307, 2010. URL https://ieeexplore.ieee.org/abstract/document/6388466/. Publisher: KICS.

Keith Ball. An Elementary Introduction to Modern Convex Geometry. In Silvio Levy (ed.), *Flavors of Geometry*, Mathematical Sciences Research Institute Publications, pp. 1–58. Cambridge University Press, Cambridge, 1997. ISBN 978-0-521-62962-1. doi: 10.1017/9781009701853.002. URL https://www.cambridge.org/core/books/flavors-of-geometry/an-elementary-introduction-to-modern-convex-geometry/09EA8B107A9ECA496914D95520C9EAFB.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory 2nd Edition*. Wiley-Interscience, Hoboken, N.J, 2006. ISBN 978-0-471-24195-9.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models, October 2023. URL http://arxiv.org/abs/2309.08600. arXiv:2309.08600.

Sanjoy Dasgupta and Anupam Gupta. An elementary proof of a theorem of Johnson and Lindenstrauss. *Random Structures & Algorithms*, 22(1):60–65, January 2003. ISSN 1042-9832, 1098-2418. doi: 10.1002/rsa.10073. URL https://onlinelibrary.wiley.com/doi/10.1002/rsa.10073.

David L. Donoho and Jared Tanner. Counting Faces of Randomly Projected Polytopes When the Projection Radically Lowers Dimension. *Journal of the American Mathematical Society*, 22(1):1–53, 2009. ISSN 0894-0347. URL https://www.jstor.org/stable/40587224. Publisher: American Mathematical Society.

David L. Donoho, Arian Maleki, and Andrea Montanari. Message-passing algorithms for compressed sensing. *Proceedings of the National Academy of Sciences*, 106(45):18914–18919, November 2009. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.0909892106. URL https://pnas.org/doi/full/10.1073/pnas.0909892106.

M. Elad. *Sparse and redundant representations: from theory to applications in signal and image processing*. Springer, New York, 2010. ISBN 978-1-4419-7010-7 978-1-4419-7011-4. OCLC: ocn646114450.

Manaal Faruqui, Yulia Tsvetkov, Dani Yogatama, Chris Dyer, and Noah A. Smith. Sparse Overcomplete Word Vector Representations. In Chengqing Zong and Michael Strube (eds.), *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1491–1500, Beijing, China, July 2015. Association for Computational Linguistics. doi: 10.3115/v1/P15-1144. URL https://aclanthology.org/P15-1144/.

Simon Foucart and Holger Rauhut. Sparse Recovery with Random Matrices. In Simon Foucart and Holger Rauhut (eds.), *A Mathematical Introduction to Compressive Sensing*, pp. 271–310. Springer, New York, NY, 2013. ISBN 978-0-8176-4948-7. doi: 10.1007/978-0-8176-4948-7_9. URL https://doi.org/10.1007/978-0-8176-4948-7_9.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, June 2024. URL http://arxiv.org/abs/2406.04093. arXiv:2406.04093 [cs].

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 35:17359–17372, 2022.

Anh Nguyen, Jason Yosinski, and Jeff Clune. Multifaceted Feature Visualization: Uncovering the Different Types of Features Learned By Each Neuron in Deep Neural Networks, May 2016. URL http://arxiv.org/abs/1602.03616. arXiv:1602.03616 [cs].

Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom In: An Introduction to Circuits. *Distill*, 5(3):10.23915/distill.00024.001, March 2020. ISSN 2476-0757. doi: 10.23915/distill.00024.001. URL https://distill.pub/2020/circuits/zoom-in.

David E. Rumelhart, James L. McClelland, and AU. *Parallel Distributed Processing: Explorations in the Microstructure of Cognition: Foundations*. The MIT Press, 1986. ISBN 978-0-262-29140-8. doi: 10.7551/mitpress/5236.001.0001.

Lee Sharkey, Bilal Chughtai, Joshua Batson, Jack Lindsey, Jeff Wu, Lucius Bushnaq, Nicholas Goldowsky-Dill, Stefan Heimersheim, Alejandro Ortega, Joseph Bloom, Stella Biderman, Adria Garriga-Alonso, Arthur Conmy, Neel Nanda, Jessica Rumbelow, Martin Wattenberg, Nandi Schoots, Joseph Miller, Eric J. Michaud, Stephen Casper, Max Tegmark, William Saunders, David Bau, Eric Todd, Atticus Geiger, Mor Geva, Jesse Hoogland, Daniel Murfet, and Tom McGrath. Open Problems in Mechanistic Interpretability, January 2025. URL http://arxiv.org/abs/2501.16496. arXiv:2501.16496 [cs].

Keigo Takeuchi. Generalized Approximate Message-Passing for Compressed Sensing with Sublinear Sparsity, September 2024. URL `https://arxiv.org/abs/2409.06320v2`.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. *Transformer Circuits Thread*, May 2024. URL `https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html`.

Simon Thorpe. Local vs. Distributed Coding. *Intellectica*, 8(2):3–40, 1989. doi: 10.3406/intel.1989.873. URL `https://www.persee.fr/doc/intel_0769-4113_1989_num_8_2_873`. Publisher: Persée - Portail des revues scientifiques en SHS.

Jian Wang, Seokbeop Kwon, and Byonghyo Shim. Generalized Orthogonal Matching Pursuit. *IEEE Transactions on Signal Processing*, 60(12):6202–6216, December 2012. ISSN 1941-0476. doi: 10.1109/TSP.2012.2218810. URL `https://ieeexplore.ieee.org/document/6302206`.

L. Welch. Lower bounds on the maximum cross correlation of signals (Corresp.). *IEEE Transactions on Information Theory*, 20(3):397–399, May 1974. ISSN 1557-9654. doi: 10.1109/TIT.1974.1055219. URL `https://ieeexplore.ieee.org/document/1055219`.

Zeyu Yun, Yubei Chen, Bruno Olshausen, and Yann LeCun. Transformer visualization via dictionary learning: contextualized embedding as a linear superposition of transformer factors. In Eneko Agirre, Marianna Apidianaki, and Ivan Vulić (eds.), *Proceedings of Deep Learning Inside Out (DeeLIO): The 2nd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pp. 1–10, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.deelio-1.1. URL `https://aclanthology.org/2021.deelio-1.1/`.

Changwan Zhang and Yue Wang. A sample survey study of poly-semantic neurons in deep CNNs. In *International Conference on Computer Graphics, Artificial Intelligence, and Data Processing (ICCAID 2022)*, volume 12604, pp. 849–855. SPIE, May 2023. doi: 10.1117/12.2674650.

A Matched Filters

Consider the problem of inferring a scalar S from the sum

$$X = Sf + Z$$

where $f \in \mathbb{R}^n$ is fixed and Z is a Gaussian variable independent from S. Suppose for simplicity that Z has non-singular covariance Σ, so that $-\ln p(z) = 1/2\|z\|_\Sigma^2$ where

$$\|z\|_\Sigma^2 = z^T \Sigma^{-1} z.$$

Then a routine calculation shows that

$$\begin{aligned} &-\ln p(S = s | X = x) \\ &= C(x) - \ln p(s) + \frac{1}{2}\left(s - \frac{\langle f, x\rangle_\Sigma}{\|f\|_\Sigma^2}\right)^2 \|f\|_\Sigma^2 \end{aligned} \tag{6}$$

where $C(x)$ is a constant depending only on x and $\langle -, - \rangle_\Sigma$ is the inner product associated with the norm $\|-\|_\Sigma$. In particular, the distribution of S conditional on X is only a function of the inner product $\langle f, X\rangle_\Sigma$. The **matched filter** for S is the linear function

$$\lambda(x) = \frac{\langle f, x\rangle_\Sigma}{\|f\|_\Sigma^2},$$

and can be understood as providing the maximum likelihood estimate for S.

In general, the quality of a linear filter λ' as an estimate for S can be measured by the signal-to-noise ratio

$$\rho(\lambda') = \frac{(\lambda'(f))^2}{\mathrm{Var}_Z\, \lambda'(Z)}.$$

The maximum possible signal-to-noise ratio is $\|f\|_\Sigma^2$, which is achieved by λ. Given an improper uniform prior on S, (6) shows the posterior on S conditional on X is a Gaussian with mean $\lambda(X)$ and precision $\|f\|_\Sigma^2$.

B Estimates for the Binomial Coefficient

To estimate $\ln \binom{N}{k}$, it is helpful to first remember the elementary inequalities

$$\left(\frac{N}{k}\right)^k \le \binom{N}{k} \le \left(\frac{eN}{k}\right)^k.$$

Taking logarithms gives $k \ln(N/k) \le \ln \binom{N}{k} \le k \ln(eN/k)$, which proves

$$\ln \binom{N}{k} = k \ln(N/k) + O(k).$$

When $k \ll N$, the upper bound $\ln \binom{N}{k} \le k \ln(eN/k)$ is a better approximation; in fact

$$\ln \binom{N}{k} = k \ln(eN/k) + O(k^2/N).$$

In terms of $k = N^\eta$ the remainder is $O(N^{2\eta-1})$, which vanishes for $\eta < 1/2$.

One heuristic justification for this finer estimate is to consider a random subset $Y \subseteq [N]$ where each element is included independently with probability $s = k/N$. Then, where

$$h(s) = -s \ln s - (1-s)\ln(1-s) = -s \ln s + s + O(s^2)$$

is the binary entropy function, the entropy of Y is

$$\begin{aligned} H(Y) = h(s)N &= -sN \ln s + sN + O(s^2 N) \\ &= k \ln(eN/k) + O(k^2/N). \end{aligned}$$

We expect that for large N the number of elements in Y concentrates sharply around k, and so $H(Y)$ should be close to $\ln \binom{N}{k}$.

Indeed this can be proven using the Stirling approximation

$$\ln n! = n \ln n - n + (1/2)\ln(2\pi n) + O(n^{-1}).$$

Substituting into the binomial gives

$$\begin{aligned} \ln \binom{N}{k} &= \ln N! - \ln k! - \ln(N-k)! \\ &= k \ln(N/k) + (N-k)\ln(1 + k/(N-k)) + O(1/N), \end{aligned}$$

and finally the Taylor approximation $\ln(1 + k/(N-k)) = k/N + O(k^2/N^2)$ yields

$$\begin{aligned} \ln \binom{N}{k} &= k \ln(N/k) + (N-k)[k/N + O(k^2/N^2)] + O(1/N) \\ &= k \ln(N/k) + k(N-k)/N + O(k^2/N) \\ &= k \ln(N/k) + k(1 - k/N) + O(k^2/N) \\ &= k \ln(N/k) + k + O(k^2/N) \\ &= k \ln(eN/k) + O(k^2/N). \end{aligned}$$

C Basic Results on Random Vectors

The following is a well-known estimate for the tail of a Gaussian.

Proposition 5. *When Z is a unit Gaussian, we have*

$$\ln \mathrm{P}(Z \geq a) = -\frac{1}{2}a^2 - \ln a - \frac{1}{2}\ln 2\pi + o(1).$$

for $a \to +\infty$.

In fact, the leading-order term $-\frac{1}{2}a^2$ is an upper bound on $\ln \mathrm{P}(Z \geq a)$. One simple way to establish such estimates on tail probabilities is via the Chernoff inequality: for any random variable X and $\lambda > 0$,

$$\mathrm{P}(X \geq a) \leq e^{-\lambda a}\,\mathrm{E}\,e^{\lambda X} = \exp(-\lambda a + K_X(\lambda))$$

where $K_X(\lambda) = \ln \mathrm{E}\,e^{\lambda X}$ is the cumulant generating function. In general, putting $\lambda = a$ in the Chernoff inequality for a random variable X with $K_X(\lambda) \leq \lambda^2/2$ gives

$$\mathrm{P}(X \geq a) \leq \exp(-a^2/2),$$

which is called a sub-Gaussian tail bound.

Let $X_1, \ldots, X_k$ be independent Rademacher random variables, each uniformly distributed on $\{-1, 1\}$. Then $K_{X_i}(\lambda) = \ln\cosh\lambda \leq \lambda^2/2$, so

$$K_{R_k}(\lambda) = \sum_{i=1}^{k} K_{X_i}(\lambda) \leq k\lambda^2/2$$

where $R_k = X_1 + \cdots + X_k$. This proves the following.

Proposition 6 (Chernoff bound for Rademacher sums)**.** *For R_k a sum of k independent Rademacher variables and $t > 0$,*

$$\mathrm{P}(R_k \geq t) \leq \exp\left(-\frac{t^2}{2k}\right).$$

This is enough to prove Proposition 1, which we restate here for convenience.

Proposition. *Let $d > 2\epsilon^{-2}(2\ln N + \ln p^{-1})$, and let*

$$\{F_1, \ldots, F_N\} \subseteq \{-1/\sqrt{d}, 1/\sqrt{d}\}^d$$

be random vectors with independent, uniformly distributed entries. Then $|\langle F_i, F_j\rangle| < \epsilon$ for all $i \neq j$ with probability at least $(1-p)$.

Proof. Each inner product $\langle F_i, F_j\rangle$ is distributed like R_d/d where R_d is a sum of d Rademacher variables. By the Chernoff bound, $\mathrm{P}(\langle F_i, F_j\rangle \geq \epsilon) = \mathrm{P}(R_d \geq d\epsilon) \leq \exp(-d\epsilon^2/2)$. By symmetry and a union bound over all $\binom{N}{2} < N^2/2$ pairs,

$$\mathrm{P}(\exists\, i \neq j : |\langle F_i, F_j\rangle| \geq \epsilon) \leq N^2 \exp(-d\epsilon^2/2).$$

This is at most p when $d \geq 2\epsilon^{-2}(2\ln N + \ln p^{-1})$. □

The interested reader should also compare this result to the Johnson-Lindenstrauss lemma, which is proved in a very similar way. (See Dasgupta & Gupta (2003) for a proof, or the last section of Foucart & Rauhut (2013) for a discussion of the JL lemma with some broader context.)

Now, consider the inner product of two d-dimensional codewords X_i, X_j drawn from a random spherical dictionary. (That is, X_i and X_j are uniformly distributed on the unit sphere in d dimensions.) Write $b(d, t)$ for the probability that $\langle X_i, X_j\rangle \geq t$. This number can also be viewed as fraction of the unit sphere occupied by the spherical cap

$$\{(v_1, \ldots, v_d) : v_1^2 + \cdots + v_d^2 = 1,\ v_1 \geq t\}$$

Using the fact that normalizing a Gaussian random vector gives a uniform draw from the unit sphere, it is possible to establish that

$$b(d,t) \le (1+o(1)) \exp\left(-\frac{t^2 d}{2}\right)$$

using Chernoff inequalities. However, for the proof of Proposition 3, we also need a *lower* bound on this tail probability. This is accomplished by the following result, which is a corollary of some ideas from Lecture 2 of Ball (1997).

Proposition 7. *For all $t \in (0,1)$,*

$$\exp\left(-\frac{t^2 d}{2(1-t^2)}\right) \le b(d,t) \le \exp\left(-\frac{t^2 d}{2}\right)$$

D Proofs of Main Results

We begin with Proposition 3, restated here for convenience. Recall that $b(d,k,N;\text{top-}k)$ is the probability that the top-k decoder fails to recover a k-sparse subset of $[N]$ from a d-dimensional superposition code.

Proposition. *Let $\epsilon > 0$ be arbitrary. Over the regime*

$$d \le (1-\epsilon) 2k \ln N, \quad 2 \le k < N/2,$$

for random spherical dictionaries it holds that

$$\lim_{N\to\infty} b(d,k,N;\textit{top-}k) = 1.$$

The proof can likely also be modified to hold for Rademacher dictionaries.

Proof. We can suppose w.l.o.g. that X is the set $\{1,\ldots,k\}$. As before, we write F for the dictionary, F_i for its constituent codewords, and $Y = FX = F_1 + \cdots + F_k$ for the superposition code.

It is easy to show that, with probability bounded below by a constant independent of (d,k), we have $\|Y\| \ge \sqrt{k}$ and $\langle F_1, Y\rangle < 1$ so long as $k > 2$.

Let us condition on this event. The remaining codewords $F_{k+1},\ldots,F_N$ are independent, uniform draws from the unit sphere. Let B_i be the event that $\langle F_i, Y\rangle \ge 1$, and let B be the union of the $(N-k)$ events $\{B_{k+1},\ldots,B_N\}$. If B holds, then top-k decoding will not correctly recover X from the code Y.

For each $i \in \{k+1,\ldots,N\}$, we have

$$\mathrm{P}(B_i) = \mathrm{P}(\langle F_i, Y\rangle \ge 1) = \mathrm{P}\left(\left\langle F_i, \frac{Y}{\|Y\|}\right\rangle \ge \frac{1}{\|Y\|}\right).$$

For each $i > k$, the pair $(Y_i, Y/\|Y\|)$ is a pair of independent, uniform draws from the unit sphere, so combining Proposition 7 with our assumption $\|Y\| \ge \sqrt{k}$ gives

$$\mathrm{P}(B_i) \ge \exp\left(-\frac{d}{2(\|Y\|^2-1)}\right) > \exp\left(-\frac{d}{2(k-1)}\right).$$

Since the events B_i are independent, we can lower bound $\mathrm{P}(B)$ by

$$\mathrm{P}(B) = \mathrm{P}\left(\bigcup_i B_i\right) = 1 - \prod_{i=k+1}^{N} (1-\mathrm{P}(B_i)) \ge 1 - \exp\left(\sum_i -\mathrm{P}(B_i)\right).$$

Substituting in our lower bound on $\mathrm{P}(B_i)$ gives

$$\mathrm{P}(B) \ge 1 - \exp\left(-(N-k)\cdot \exp\left(-\frac{d}{2(k-1)}\right)\right).$$

When $d \leq (1-\epsilon)2k \ln N$, we obtain

$$\begin{aligned} \mathrm{P}(B) &\geq 1 - \exp\left(-(N-k)\exp\left(-\frac{(1-\epsilon)k\ln N}{k-1}\right)\right) \\ &\geq 1 - \exp\left(-(N-k)\exp(-(1-\epsilon)\ln N)\right). \end{aligned}$$

Finally, by our assumptions $k < N/2$ and $\epsilon > 0$, we conclude that

$$b(d,k,N;\text{top-}k) \geq P(B) \geq 1 - \exp\left(-\frac{1}{2}\exp(\epsilon \ln N)\right) \xrightarrow{N\to\infty} 1.$$

□

Now we turn to the proof of Proposition 4, restated below. Recall that $b(d,k,N;\tau)$ is the failure probability for threshold decoding at level τ.

Proposition. *For $\eta \in (0,1)$, let $k \leq N^\eta$. Then for any $\epsilon > 0$, over the regime*

$$d \geq (1+\epsilon)2(1+\sqrt{\eta})^2 k \ln N,$$

for Rademacher dictionaries it holds that

$$\lim_{N\to\infty} b(d,k,N;\tau(\eta)) = 0$$

where $\tau(\eta) = (1+\sqrt{\eta})^{-1}$.

Our proof relies in the following bound on $b(d,k,N;\tau)$, valid for Rademacher dictionaries. Similar bounds can be proved for other random dictionaries.

Lemma 1. *For all $d,k,N \in \mathbb{N}$ and $\tau \in (0,1)$,*

$$b(d,k,N;\tau) \leq k\exp\left(-\frac{(1-\tau)^2 d}{2(k-1)}\right) + (N-k)\exp\left(-\frac{\tau^2 d}{2k}\right).$$

Proof. Suppose w.l.o.g. that $X = \{1,\ldots,k\}$ and let $Y = FX = F_1 + \cdots + F_k$.

Define the families of events

$$\begin{aligned} A_i &= [\langle F_i, Y\rangle < \tau] \quad \text{for } 1 \leq i \leq k, \\ B_i &= [\langle F_i, Y\rangle \geq \tau] \quad \text{for } i > k. \end{aligned}$$

Let R_t be a sum of t independent Rademacher variables, as in Proposition 6. For $i > k$, $\langle F_i, Y\rangle$ is distributed as R_{kd}/d. Thus, by a Chernoff bound,

$$\mathrm{P}(B_i) = \mathrm{P}(\langle F_i, Y\rangle \geq \tau) = \mathrm{P}(R_{kd} \geq d\tau) \leq \exp\left(-\frac{\tau^2 d}{2k}\right).$$

Similarly, for $i \leq k$, $\langle F_i, Y\rangle = \langle F_i, F_i\rangle + \langle F_i, \sum_{j\neq i} F_j\rangle$ is distributed like $1 + R_{(k-1)d}/d$, and so

$$\begin{aligned} \mathrm{P}(A_i) &= \mathrm{P}(\langle F_i, Y\rangle < \tau) = \mathrm{P}(R_{(k-1)d} < (\tau-1)d) \\ &\leq \mathrm{P}(R_{(k-1)d} \geq (1-\tau)d) \leq \exp\left(-\frac{(1-\tau)^2 d}{2(k-1)}\right). \end{aligned}$$

Our conclusion follows from a union bound

$$b(d,k,N;\tau) = \mathrm{P}\left(\bigcup_i A_i \cup \bigcup_i B_i\right) \leq \sum_{i=1}^{k} \mathrm{P}(A_i) + \sum_{i=k+1}^{N} \mathrm{P}(B_i).$$

□

We are now prepared to complete the proof.

Proof of Proposition 4. Put $d = Ck \ln N$ for some constant C to be determined. Using the assumption $k \leq N^\eta$ and the previous lemma lets us bound $b(d, k, N; \tau)$ by a sum of powers of N with exponents depending only on η and C:

$$\begin{aligned} b(d, k, N; \tau) &\leq N^\eta \exp\left(-\frac{(1-\tau)^2 Ck \ln N}{2k}\right) + N \exp\left(-\frac{\tau^2 Ck \ln N}{2k}\right) \\ &= \exp\left(\left(\eta - \frac{(1-\tau)^2 C}{2}\right) \ln N\right) + \exp\left(\left(1 - \frac{\tau^2 C}{2}\right) \ln N\right). \end{aligned} \tag{7}$$

Putting $\tau = (1 + \sqrt{\eta})^{-1}$,

$$(1-\tau)^2 = \frac{\eta}{(1+\sqrt{\eta})^2} \quad \text{and} \quad \tau^2 = \frac{1}{(1+\sqrt{\eta})^2}$$

and so

$$b(d, k, N; \tau) \leq \exp\left(\left(\eta - \frac{\eta C}{2(1+\sqrt{\eta})^2}\right) \ln N\right) + \exp\left(\left(1 - \frac{C}{2(1+\sqrt{\eta})^2}\right) \ln N\right).$$

As $N \to \infty$, both terms vanish for any fixed $C > 2(1 + \sqrt{\eta})^2$, which completes the proof. □

In the proof above, the choice of threshold $\tau = (1 + \sqrt{\eta})^{-1}$ is only justified post hoc. We now explain how to derive this expression and how it relates to the MAP threshold derived in Section 1

For a given η, we're interested in knowing the critical value of C above for which some choice of threshold τ allows both terms in (7) to converge to 0, meaning that both exponents are negative:

$$\min_{\tau \in (0,1)} \max \left\{ \eta - \frac{(1-\tau)^2 C}{2}, 1 - \frac{\tau^2 C}{2} \right\} < 0.$$

The minimizer in τ is $\tau = 1/2 + (1 - \eta)/C$, at which point both exponents are

$$\frac{1+\eta}{2} - \frac{C}{8} - \frac{(1-\eta)^2}{2C}.$$

Solving a quadratic shows that this expression is negative so long as $C > 2(1 + \sqrt{\eta})^2$. Substituting this critical value into our expression for the threshold τ achiving the best exponents in the upper bound and simplifying gives $\tau = (1 + \sqrt{\eta})^{-1}$.

Finally, note that the threshold minimizing the exponent of the Chernoff-type bounds in this proof agrees with the threshold we derived from a maximum a posteriori estimate in Section 1 (Equation (5)). Substituting $C = d/(k \ln N)$ and $\eta = \ln k / \ln N$ gives

$$\tau_{\text{Chernoff}} = \frac{1}{2} + \frac{1-\eta}{C} = \frac{1}{2} + \frac{k \ln N}{d} - \frac{k \ln k}{d} = \frac{1}{2} - \frac{k}{d} \ln \frac{k}{N}$$

while, following our earlier discussion, the maximum a posteriori threshold is

$$\tau_{\text{MAP}} = \frac{1}{2} - \rho^{-1} \ln \frac{\mathrm{P}(X_i = 1)}{\mathrm{P}(X_i = 0)} = \frac{1}{2} - \frac{k}{d} \ln \frac{k}{N} + O\left(\frac{k}{N}\right)$$

when $k/N \to 0$.